

Raul-Andrei Pop

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Education

École Polytechnique, Paris

Sep 2024 – Jul 2027

B.Sc. in Mathematics and Computer Science — current GPA: 4.0/4.0

Class of 2027

- **Student Representative:** active member of the Academic Council, advocating for academic improvements.
- Working with the Computer Science department to improve the grading system and other academic aspects.
- Organizing the half-semester course review and presenting student feedback to professors.
- **Coursework:** Advanced Python, C++ OOP, Rocq Prover, Linear Algebra I & II, Probability & Statistics, Econometrics.

International Computer High School of Bucharest — Computer Science (Olympiad track), GPA: 9.99/10

Class of 2024

Additional Training & Learning

- **Jane Street** First-Year Trading and Technology Program (2025).
- **Susquehanna** Technology Discovery Program (2026).
- **Stanford** Supervised Machine Learning — online course, Certificate of Completion.
- **JetBrains** Algorithms and Code Training School (2024).
- Osijek Competitive Programming Camp (Winter 2024, Osijek; Summer 2025, Bucharest).

Work Experience

Databricks, Berlin — Software Engineering Intern

Jul – Sep 2026

Helbako Electronica SRL — Software Developer

Jun 2023 – Feb 2024

- Created Jenkins pipelines using Groovy scripts; standardized project deployment.
- Developed Python scripts to automate file handling.
- Gained experience with AUTOSAR and MISRA coding standards.

École Polytechnique — Bachelor Math Committee (BMX)

Sep 2025 – Present

Leadership Board — Competitions Organiser

- Developed a local server that graded and displayed live standings for participants of our Estimathon.
- Helped organise seminars with 100+ attendees.
- Organized École Polytechnique's Integration Bee.

Teaching

Polytechnic University of Timișoara & Upper School

2023 & 2024–2025

Algorithms Course Teacher (in partnership with the University)

- Organised and taught the algorithms summer course (7 sessions, 4 hours each; 2023 and 2024).
- Helped organise an ICPC practice contest.
- Taught graph theory lessons for an advanced algorithms class of 1st-year university students.

Personal Projects

Quantification of measurement incompatibility of MUBs [\[work on GitHub\]](#)

Lab Research Project, École Polytechnique, 2025

- Studied the incompatibility of Mutually Unbiased Bases (MUBs) in quantum mechanics.
- Reproduced numerical results for maximal robustness for Joint Measurability of p MUBs
- Contributed a new framework for quantifying more general incompatibility using SDPs on measurement hypergraphs
- Derived analytical lower bound for pairwise incompatibility case, which holds up to machine precision for the cases tested

Travel Reunion Planner

HackUPC 2025

- Tool that finds the most efficient set of flights for several people to meet, based on configurable factors.
- Integrated an LLM to interact with users and compute sentiment-analysis-based preferences.

Analyzed grokking on MLPs and Transformers for poisoned datasets [\[work on GitHub\]](#)

ML project, Ecole Polytechnique, 2026

- Investigated the phenomenon of grokking in models trained or fine-tuned on poisoned datasets.
- Analyzed the robustness of MLPs and Transformers under adversarial attacks.
- Concluded that more complex models (Transformers) are more prone to generalize structured poisoning in **small** fine-tuning sets

Chess Engine [\[work on GitHub\]](#)

C++ Semester Project, École Polytechnique, 2025-2026

- Developed a chess engine in C++ as a semester project for an Object-Oriented Programming class.
- The bot reaches 2100 ELO and is playable on Lichess.

Competitive Programming & Mathematics

- **ICPC SWERC** (Southwestern Europe Regional Contest): 17th place (Nov 2025), 21st place (Dec 2024).
- **Romanian Artificial Intelligence Olympiad** — Top 20 (Jul 2024).
- **National Informatics Olympiad** — Gold Medal (Apr 2024).
- **International Zhautykov Olympiad** — Silver Medal, Informatics (Jan 2024).
- **International Informatics Olympiad in Teams** — Gold Medal / 1st Prize (May 2023).
- **National Mathematics Olympiad** — Bronze Medal (Apr 2022).
- **Codeforces:** max rating 2143 (Master). **Meta Hacker Cup** 2023 & 2024 — qualified for Round 2.

Technical Skills

Languages & Libraries: Python (NumPy, Pandas, PyTorch, Matplotlib, Asyncio), C++ (STL, OOP, DSA, templates), Java, LaTeX.

Sports

National Karting Champion — 2013, 2017 and 2019.